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Signed: _____Leyre_____

Date: _____02/04/2025_____

Link to Repo

<https://github.com/Kornii-Kuvaldin/SD2-Game>

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Game Details

Summary of the Game

Subterra Showdown is a 2D, 2-player game. Two players race each other to see who can make the most money in a limited time. To gain the money, the player will need to mine certain precious gems that they can find scattered throughout the underground.

Split Screen

This is a two-player game where two players race each other to get the most points, by mining ore. It would have been fun if the two players were on the same screen and raced each other while trying to disturb the other player, but that could be done further down the line. Right now, we have decided that splitting the screen and that each player has their own screen to mine from is a better idea.

Challenges

There were a lot of challenges while trying to implement the split screen. The main challenge was with boundaries. Player 1's boundaries were easy to make; the challenge came with the boundaries for Player 2. While Player 1's boundaries were established with the camera fixation, Player 2's were not. That led to the player not showing on screen or not moving.

This was easily solved by manipulating the X-axis on Player 2 and creating a method to set the boundaries for the players in the Player class.

Countdown Timer

As mentioned in the previous review, we have finally settled on actually changing the times display to show it in minutes and seconds instead of just seconds like it did in the last code review.

Restart

This was a problem that has been carried over since the first code review. In this code review, we finally fixed the issue by adding a new array that rewrites itself every time it restarts. The restart method also sets the timer back to 2 minutes, sets the players to their initial positions, and repaints every block that was broken in the previous game.

Challenges

The restart was being avoided by a code that we implemented to prevent the game from restarting, because every time the game restarted, it crashed due to it trying to rewrite or delete the array while it was still running.

Thanks to the avoidance, the countdown timer did not work properly before we got the restart method to work; the timer went into the negatives since it did not restart. Once the method was fixed, the timer could be correctly implemented.

Lag

The ore generation was still causing some lag due to the fact that it was painting more blocks than the screen actually showed. This was solved by trial and error to show what was the minimum number of rows and columns that needed to be added manually.

Challenges

The main challenge of this was that the block height and width could not be used on this loop, since we still had not found a way to erase the blank space that stays with the image. So it needed to be manually modified so that it printed the actual number of needed columns and rows.

This means a lot of trials to see which one is the appropriate extra rows and columns that are needed to fill the screen without painting extra.

Mining

Ore Generation

Sprite Updates

Previously, each player only had a single sprite, and it would never update. We decided that it would make for a better viewing experience if we added different sprites that would dynamically update depending on the direction that the player was moving towards.

Challenges

The main challenge in regards to this implementation was properly loading all of the sprites and finding a way to dynamically know when to use each. It took a couple of tries, but we

ultimately settled on having multiple fields, each holding the specific sprite, and then setting the image to the related sprite when moving.

Store

Upon adding the split screen, the store had to be changed so that it would appear at the center of both screens.

Challenges

To fix this, we had to first calculate the starting point (the 0X axis) for both screens and then subtract that from the X axis of the store itself, resulting in the new proper X value for the store.

What Did Everyone Do

Leyre Raga Mosteiro

Split Screen, Modify the Timer, Solve The Problem With the Restart Method, and Reduce Lag

Lucas Rodrigues Crema

Sprite Updates, fixed the store in relation to the split screen

Kornii Kuvaldin

Mining and Ore Generation

Artem Dogadailo

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